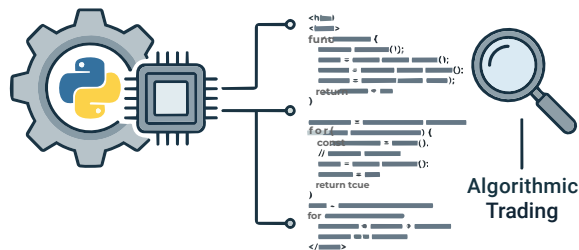


Moving Average Crossover Strategy: Algorithmic Trading with Python

Traders on the stock market can take advantage of the Moving Average Crossover Strategy to maximise the returns on their investments.



The Moving Average Crossover Strategy is probably the easiest and most effective trading algorithm that can help teach algorithmic trading. It uses only the price history of a stock and several lines of Python to cover nearly all the concepts used when designing a trading Python script strategy: market data, technical indicators, trading signals, positions, profits, risks, transaction costs, and performance. The main idea behind this strategy is to make use of the correlation between short-term and long-term moving averages to determine changes in market trends.

What is a moving average?

The moving average smoothens out the fluctuations in the prices of a stock by calculating the average price for 'n' periods preceding the current one. The formula used to compute the SMA (simple moving average) is:

$$SMA_t = \frac{P_t + P_{t-1} + \dots + P_{t-n+1}}{n}$$

...where P_t represents the closing price at time 't'. For

example, if the closing prices for five days are:

Day 1	100
Day 2	102
Day 3	101
Day 4	105
Day 5	107

...the 5-day moving average is:

$$SMA_5 = \frac{100 + 102 + 101 + 105 + 107}{5} = 103$$

With each new price, the old data point is dropped, and the new data point is added. This process can be carried out quite easily in Python using the Pandas' rolling function; a 20-day moving average will be computed as follows:

```
df["SMA20"] = df["Close"].rolling(20).mean()
```

...where 'df' is a historical market DataFrame.